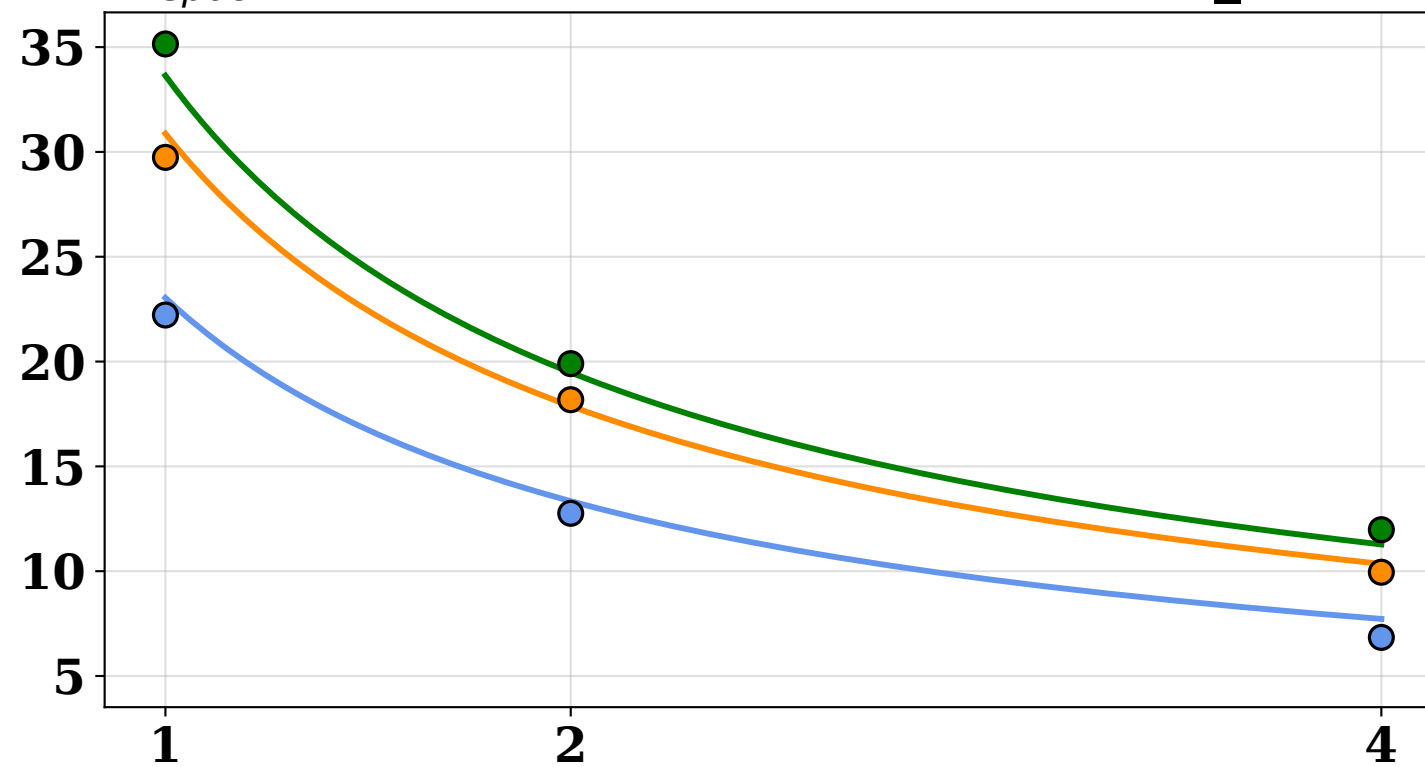


HPO runtime per epoch — power-law fit

Runtime per epoch (seconds)

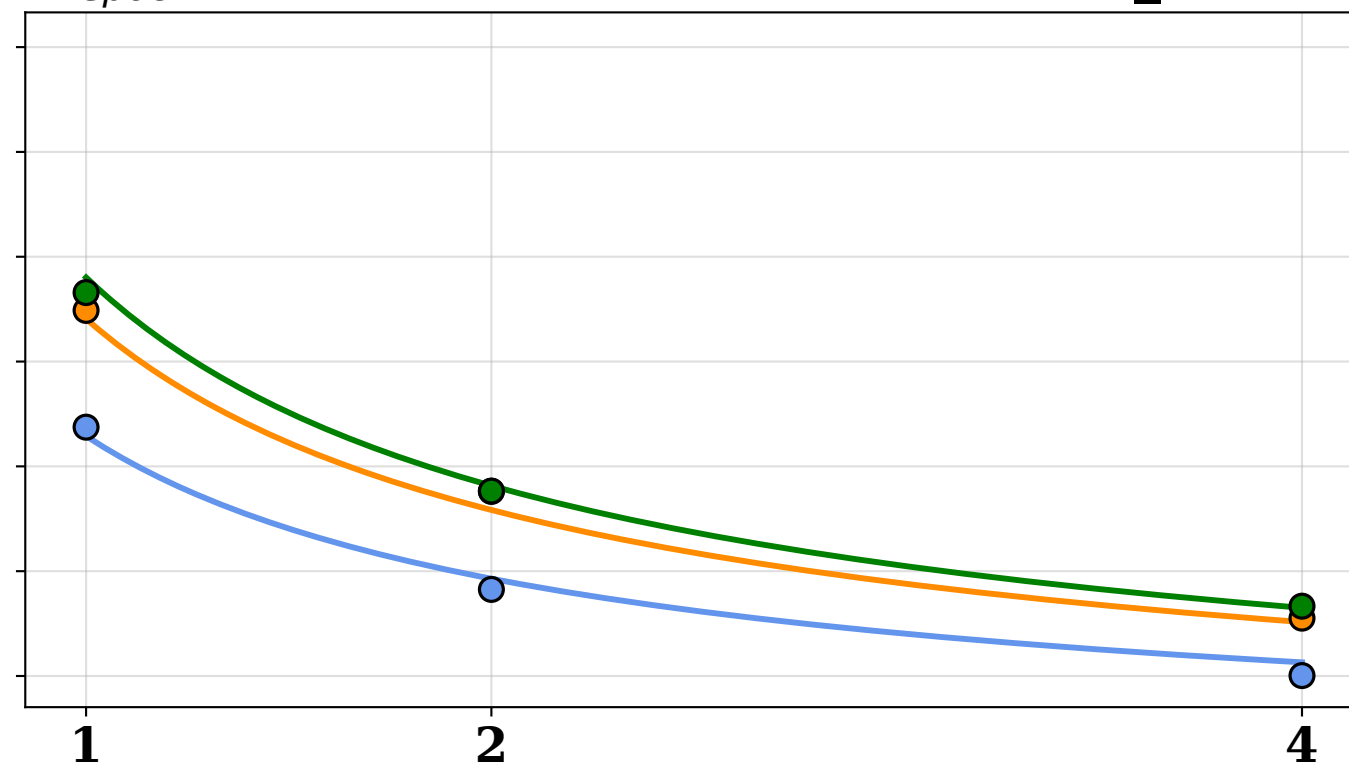
g4dn.xlarge (Tesla T4) ($R^2 = 0.9911$)

$$T_{epoch} = 23.040 \cdot \text{workers}^{-0.788} \cdot \text{model_factor}$$



g5.xlarge (NVIDIA A10G) ($R^2 = 0.9928$)

$$T_{epoch} = 16.446 \cdot \text{workers}^{-0.771} \cdot \text{model_factor}$$



Number of workers (GPUs)

— VGG19 (factor = 1.0) — Wide ResNet-101-2 (factor = 1.34) — ConvNeXt-Large (factor = 1.46)